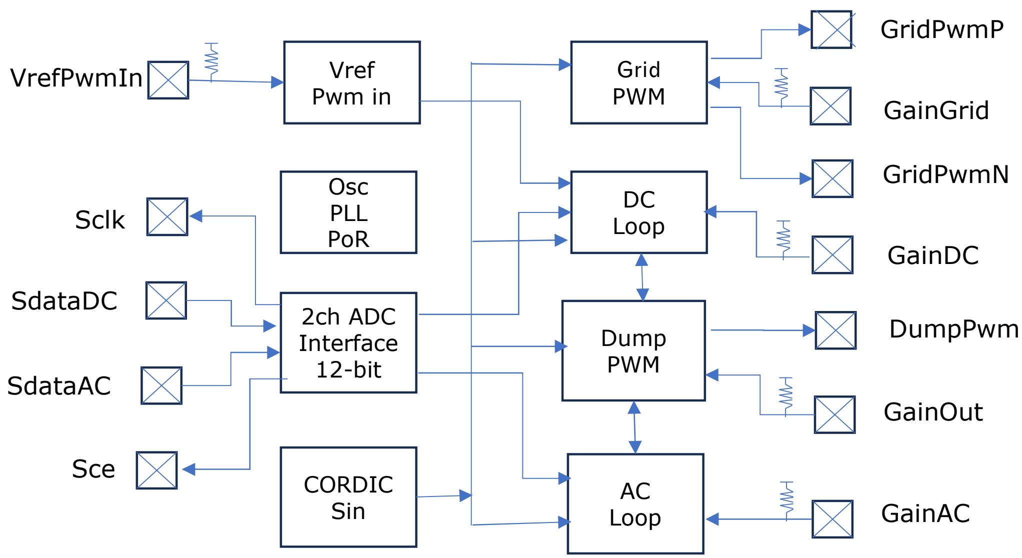
XS **Xyzzy Semiconductor**

Grid Power Controller

XS-GPC-01-xxx-xx
Architecture Draft

Block Diagram



Typical Applications

•**Micro-grid formation and stabilization** for inverter-assist and off-grid backup systems
•**Energy-balancing controllers** using dump-load regulation for solar, storage, and hybrid sources
•**Compact power-processing modules** requiring precise AC synthesis and fast DC-link control

Purpose

- Ultra-compact controllers delivering stable micro-grid formation, energy balancing, and dump-load regulation.

Features

- 48 MHz fully synchronous digital core
- Dual 3 MHz differential ADC inputs with isolation-friendly front-end
- Integrated AC-loop controller with CORDIC-based sine synthesis
- Integrated DC-loop controller with energy-balance regulation
- Deterministic timing — no firmware, no jitter, no interrupts
- On-chip PWM generation for H-bridge and dump-load control
- Two-wire directional PWM interface for external FET drivers
- Programmable Vref via single-pin PWM input
- Low-latency error filtering and accumulator-based control loops
- Built-in dump-load minimum-pulse enforcement
- Dead-time-safe PWM outputs for discrete gate drivers
- Compact 3 mm × 3 mm package for dense power modules
- Low-voltage, low-power digital architecture
- Clean integration path for isolated voltage and current sensing
- Stable operation across wide input and load conditions
- Simple external component count — no MCU required
- Fully deterministic startup and shutdown behavior
- Compatible with micro-inverter assist and inverter-to-inverter sync
- Ideal for off-grid, backup, and energy-harvesting applications
- Designed for robust operation in noisy power environments

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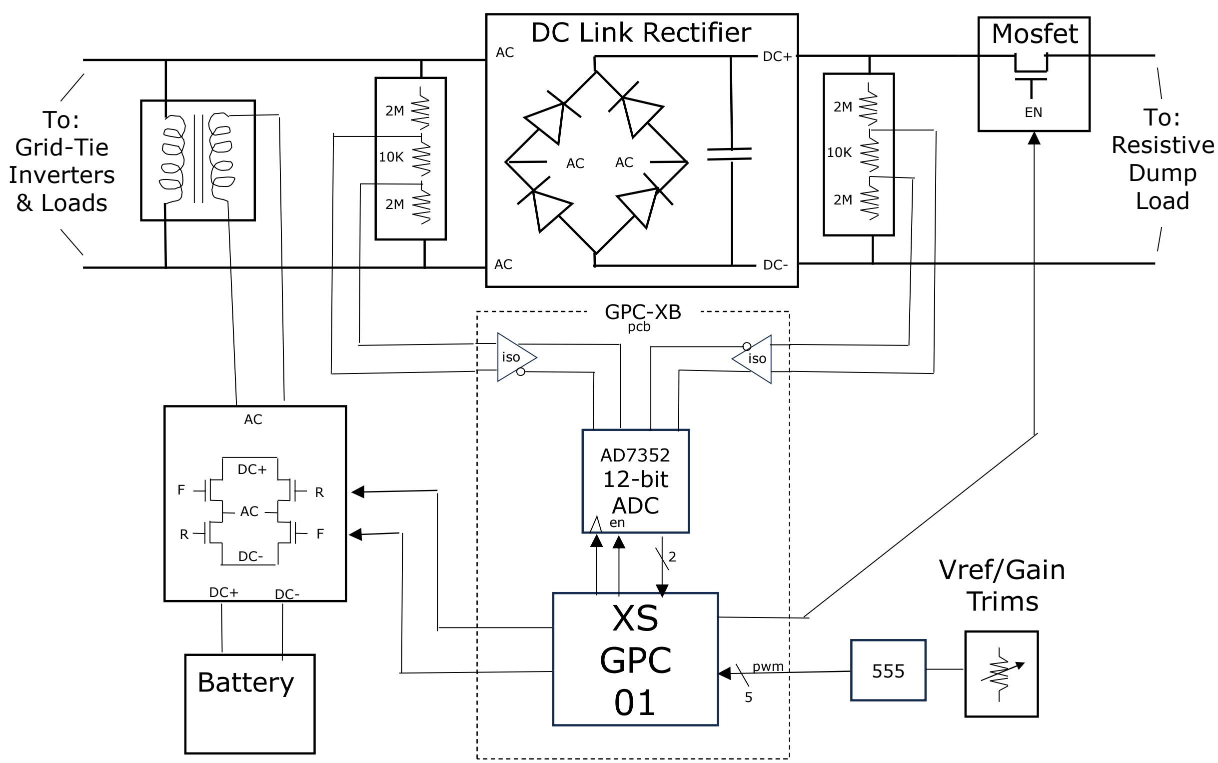
System Diagram

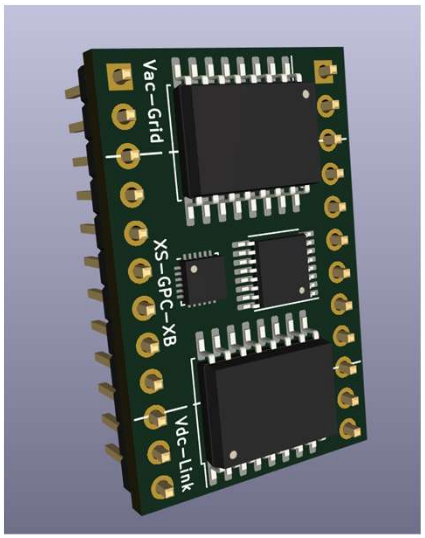


Description

The XS-GPC-01 sits at the center of a compact, low-voltage power-control system designed to synthesize a clean AC waveform while actively regulating the DC-link that feeds it. Differential isolated ADC inputs provide real-time sensing of both the AC output and the DC-link voltage, enabling the chip's dual-loop architecture to maintain stable operation across rapidly changing load and source conditions. An external H-bridge module, driven by the GPC-01's deterministic PWM outputs, generates the final AC waveform, while a dump-load MOSFET path absorbs excess energy to keep the DC-link within safe limits.

Around the GPC-01, the system integrates a small VFD-style rectifier and DC-link capacitor bank, isolated voltage and current sensors, and simple configuration elements such as potentiometers or strap pins for loop tuning. This architecture forms a complete micro-grid control core: the AC loop shapes the sine wave, the DC loop balances energy flow, and the dump-load path ensures stability during surplus generation. The result is a tightly integrated, minimal-component power-control platform suitable for experimentation, inverter-assist development, and compact energy-processing modules.

Experiment Board



Typical Applications

•**Micro-grid experimentation** using low-voltage AC synthesis and DC-link control
•**Solar-inverter behavior studies** without high-voltage hardware
•**Rapid prototyping** of dump-load regulation, inverter-assist, and energy-balancing concepts

Purpose

A compact evaluation and experimentation board for engineers and solar enthusiasts to explore, prototype, and validate the XS-GPC-01 power-control architecture in a safe, low-voltage environment.

Features

•Integrates the XS-GPC-01 power-control ASIC
•Fully low-voltage design — safe for bench-top experimentation
•Differential ADC front-end brought out for external isolated sensors
•Two-wire directional PWM outputs for H-bridge driver evaluation
•Dedicated dump-load PWM output with minimum-pulse enforcement
•On-board Vref PWM generator with trimpot for loop tuning
•Breakout headers for all primary control and sense signals
•Independent AC and DC loop visibility for debugging and tuning
•Clean separation of analog sensing and digital control domains
•Compact board layout optimized for signal integrity
•Test points for AC error, DC error, accumulators, and dump-command paths
•Compatible with discrete MOSFET drivers and small H-bridge modules
•Supports rapid iteration of loop-gain and configuration strap settings
•Designed for stable operation with small bench-top power supplies
•Ideal for validating micro-inverter startup and shutdown behavior
•Provides a safe platform for exploring energy-balance algorithms
•No firmware required — deterministic hardware control loops
•Easy integration with oscilloscopes and logic analyzers
•Suitable for educational demonstrations of AC/DC control theory
•3 mm XS-GPC-01 package showcased in a practical reference design